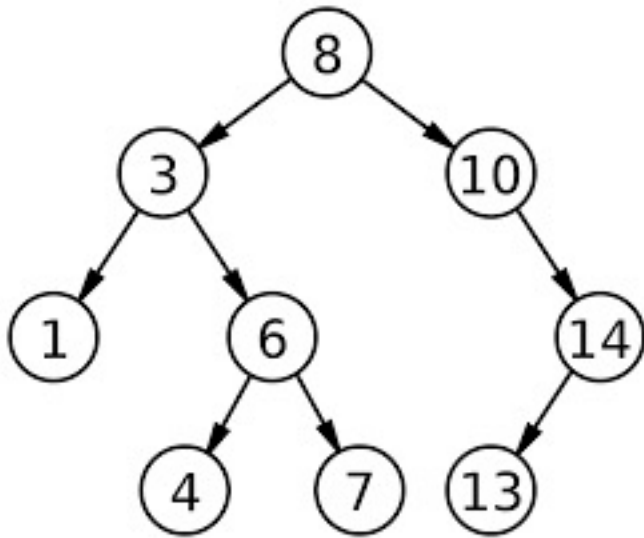


Recursive Function

(Binary Search Tree - **BST**)

What is BST?



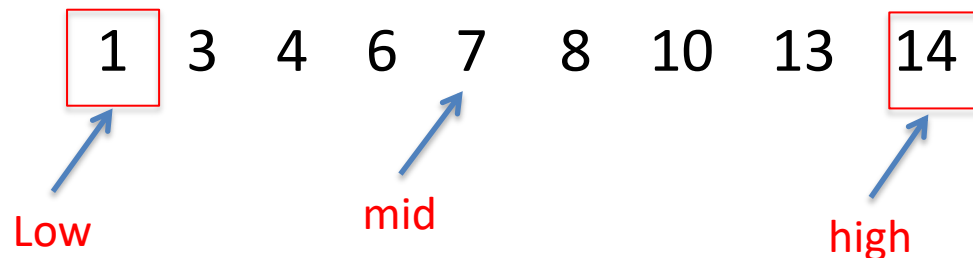
- Binary search tree (BST)** is a tree in which-
- each node has a maximum of two nodes
 - any value in the left subtree is less than the value of the node, and
 - Any value in the right subtree is higher than or equal to the value of the node.

$mid =$

1 3 4 6 7 8 10 13 14

Here, **low index = 0** and **high index = 8**

$mid = (low + high)/2$

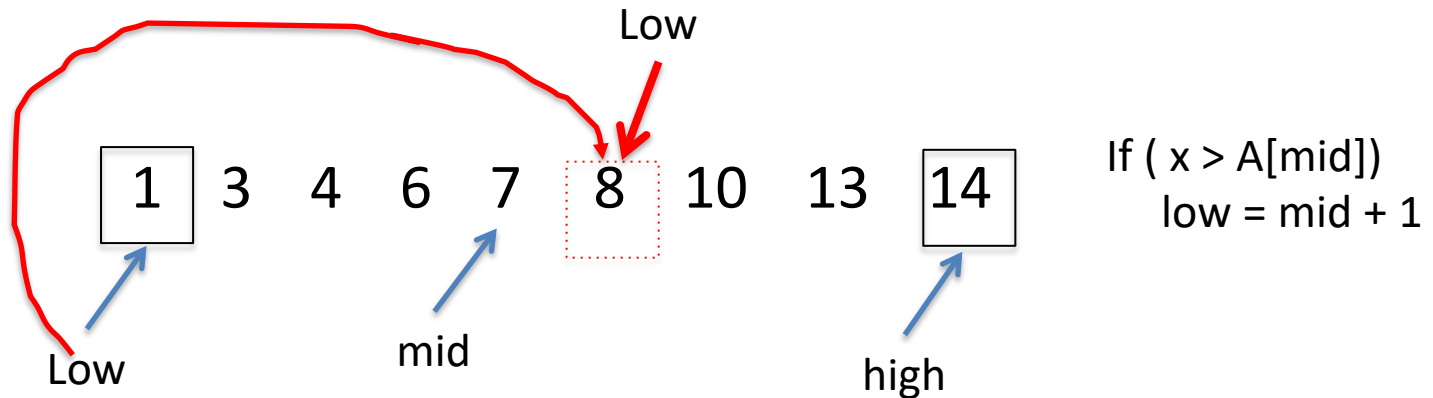


Binary Search

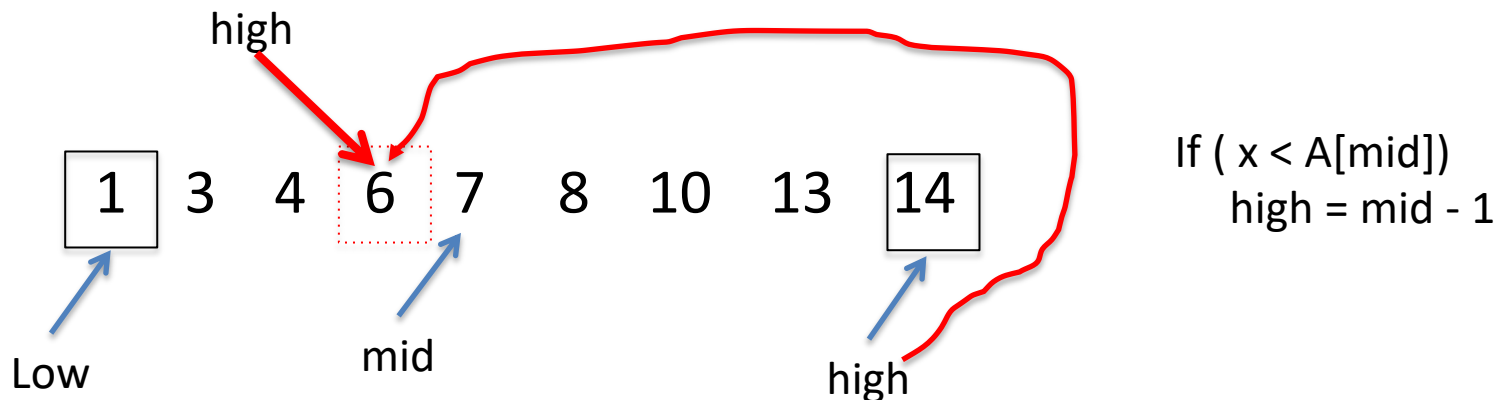
When searching for 7

If ($x == A[mid]$) return mid;

When searching for 10



When searching for 4



Iterative Function for Binary Search

```
int BinSearch(int low, int high, int x){  
    while (low < high){  
        mid = (low + high)/2;  
        if ( x == A[mid]) return mid;  
        if ( x < A[mid]) high = mid -1;  
        if ( x > A[mid]) low = mid + 1;  
    }  
    return -1;  
}
```

Recursive Function for Binary Search

Recursive Definition:

1. Termination,

- (i) if ($x == A[mid]$) return mid;
- (ii) if ($high < low$) return -1;

2. Reduction.

- (i) if ($x < A[mid]$) return BinSearch(low, mid-1, x);
- (ii) if ($x > A[mid]$) return BinSearch(mid+1, high, x);

```
int BinSearch(int low, int high, int x){  
    if ( high < low) return -1;  
    mid = (low + high)/2;  
    if ( x == A[mid]) return mid;  
    if ( x < A[mid]) BinSearch(low, mid-1, x);  
    if ( x > A[mid]) BinSearch(mid+1, high, x);  
}
```